

Acquisition Is Not Closure: Fail-Closed Control for Open-World Scientific-Literature Discovery

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Abstract

Scientific-literature discovery combines evidence acquisition with decisions about when a search may stop. These functions are often evaluated as one policy, although a locally exhausted or unavailable retrieval route does not establish that the scientific task is complete. We introduce a fail-closed control layer that separates route stopping from task closure, derives route independence from backend and query provenance, distinguishes content identity from question-conditioned processing state, and retains unresolved material routes as open obligations. A 390-task complete-gold controlled index shows the intended mechanism behavior: the governed controller reaches mean recall 0.979487 with no premature task closure, but the pre-registered comparison is underpowered and does not establish superiority. An exact-contract battery isolates the unresolved-route rule on 400 protected cases. The fail-closed controller is correct on 400 cases, compared with 250 for a donor-complete product that excludes unavailable material routes; an information-equivalent product ties the controller. The registered external Text REtrieval Conference COVID-19 (TREC-COVID) gate fails. Against a BM25 lexical baseline, recall@100 is lower by 0.01769 with a 95% bootstrap interval of $[-0.02729, -0.00906]$, and the controller reads 175.7% more candidates. Although normalized discounted cumulative gain at rank 10 (nDCG@10) is higher by 0.1488, that secondary result cannot replace the registered recall-and-cost criteria. Two later BEIR diagnostics leave all five ArguAna stopping decisions unchanged after density normalization and then query-conditional estimation, localizing the defect to a signal-inert rank-overlap marginal rather than its threshold. The contribution is therefore a bounded control method for representing acquisition authority, not evidence of external retrieval superiority or open-world completeness.

Keywords: scientific literature discovery; information retrieval; systematic review; stopping rules; coverage; open-world search; fail-closed control.

1 Introduction

Scientific-literature discovery systems increasingly combine lexical and semantic retrieval, query reformulation, citation expansion, selective reading, screening, and learned stopping. Better acquisition can recover more useful material, but it cannot answer a different question by itself: what does the observed search state authorize a system to claim about completeness? A route may be locally exhausted while another route remains untried. A provider may be unavailable rather than empty. Two differently named routes may share the same backend and query derivation. A document may have been encountered before but not processed for the current scientific question. Treating these cases as ordinary negative evidence creates false closure.

This paper studies that control boundary rather than proposing a new general retriever. The method separates an acquisition policy, which selects the next search or reading action, from a closure authority, which decides whether the current evidence state permits route stopping or task closure. Four design elements implement the separation. First, route independence is supported by backend, query-derivation, and capture provenance rather than route labels. Second, route-local stopping never certifies task-global completeness. Third, unavailable, censored, or provider-invalid material routes remain explicit open obligations. Fourth, content identity is recorded separately from question-conditioned read state, preventing duplicate discovery from inflating coverage without suppressing legitimate rereading.

The empirical question is deliberately narrow: do these control elements behave in their intended directions under complete denominators and protected exact contracts, and do they support an external superiority claim? We use a complete-gold controlled index to expose premature closure and mechanism failures, then an exact-contract battery to isolate unresolved-route authority. The external decision is supplied by a preregistered TREC-COVID comparison. Its recall and cost gates both fail. We retain the favorable nDCG result as a secondary observation, but do not use it to rescue the registered decision.

The resulting contribution is a methods and critical system-design claim. It does not claim that finite open-world search proves completeness, that the controller outperforms established retrieval systems, or that existing retrieval, screening, stopping, and memory components are novel. Instead, it provides a composable representation of what acquisition outcomes are allowed to certify, together with controlled evidence and an adverse external test that fix the present boundary.

2 Acquisition is not closure

The central design move in this paper is to separate two problems that are often collapsed into one policy. An *acquisition policy* decides what to search, retrieve, inspect, expand or reread next. A *closure authority* decides whether the evidence state is allowed to support route-local stopping or scientific task completion. Better acquisition can improve recall and reduce cost, but it does not automatically acquire authority to declare completeness. This separation makes strong external retrieval and stopping mechanisms composable rather than adversarial to the contribution of this paper.

Let x_t denote the observable search state at step t . We write an acquisition mechanism as

$$a_t \sim A(x_t),$$

where a_t may be a lexical query, a reasoning-aware query, a fielded Boolean filter, a graph-expansion action, a selective fetch, a reranking operation, or a request for another retrieval round. No novelty is claimed here for the internal form of A . Instead, acquired evidence updates both an evidence state E_t and a typed obligation state O_t . Closure is a distinct map

$$C(E_t, O_t) \in \{\text{CONTINUE, ROUTE-STOP, OPEN, CANNOT-CHECK, TASK-CLOSE}\}.$$

A utility, relevance, confidence, sufficiency or coverage signal may influence A and may recommend a local stop, but it cannot by itself delete an unresolved obligation in O_t . This interface is where the residual contribution of this paper sits.

2.1 Absorbing strong acquisition mechanisms

The acquisition layer is intentionally donor-composable. SAGE shows that keyword-oriented agent queries can make strong lexical retrieval difficult to beat and that corpus-side metadata/keyword

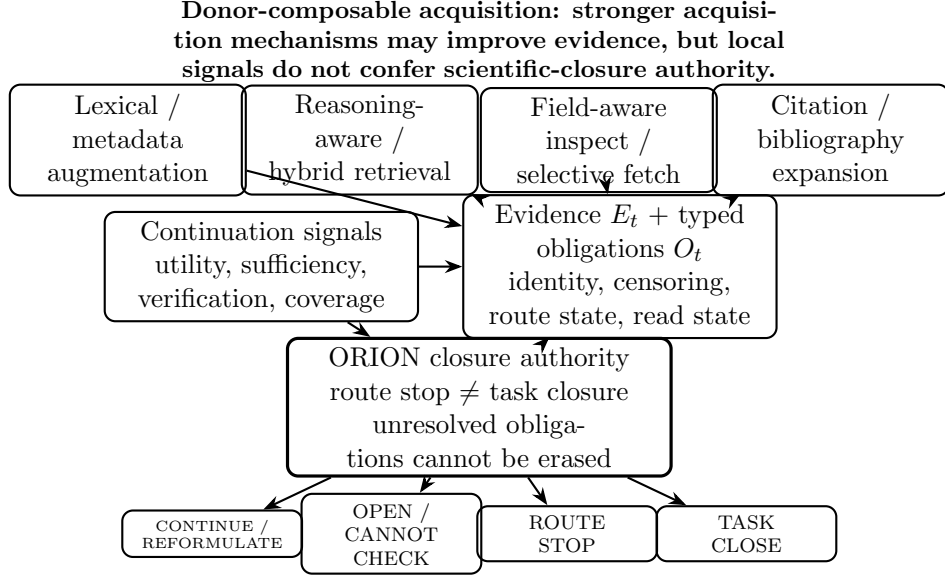


Figure 1: Acquisition mechanisms are intentionally composable: lexical/metadata augmentation, reasoning-aware or hybrid retrieval, field-aware selective fetching, citation expansion and continuation signals may all improve what is acquired. Their outputs update evidence and typed obligations, but a separate authority layer controls route stop, open and undetermined states and task closure. The figure therefore locates the residual at the meaning of acquisition outcomes rather than at a claim to have invented the strongest retriever.

augmentation can further improve retrieval [9]. AgentIR shows that an agent’s reasoning trace can be used jointly with its query as a retrieval signal [3]. SIEVE shows that fielded Boolean admission, structure-rich inspection and selective section fetching can improve the accuracy–context tradeoff [23]. Breadth-first bibliography expansion can recover literature missed by API-only retrieval, while also demonstrating why citation-list recall cannot be the only evaluation axis [20]. Lexical, metadata-augmented, reasoning-aware, field-aware and citation-expansion retrieval are therefore treated as candidate strengths to compose, not as inventions of this work.

The same rule applies to continuation and stopping signals. DeepControl uses marginal information utility to regulate continued retrieval [26]; decision-theoretic policies optimize screening stops against explicit costs and payoffs [8]; HALT uses claim–evidence coverage to verify whether expected evidence has been obtained [18]; MiCP applies conformal prediction to adaptive multi-turn reasoning to maintain a target coverage guarantee while permitting early stopping [29]; and structured Search-R1 stopping judgments explicitly encode sufficiency and remaining gaps [13]. These signals and guarantees can all be useful inputs to acquisition or route-local stopping. The question asked here is narrower: what are those signals permitted to certify when routes are dependent, unavailable, censored or only locally exhausted?

2.2 The authority residual

This decomposition yields four residual design claims that are separable from the mechanics above. First, route independence must be earned from backend, query-derivation and capture identity rather than from distinct tool names. Second, route-local stopping and task-global closure are different authority levels. Third, an unavailable or censored route remains a typed open obligation

rather than becoming an empty result. Fourth, content identity and question-conditioned processing state must remain distinct so duplicate rediscovery does not inflate coverage while a genuinely changed scientific question can trigger justified rereading.

The resulting design is deliberately asymmetric. It is permissive about *how* evidence is acquired: a stronger retriever, query generator, expansion rule or verification signal can replace a weaker one. It is conservative about *what acquisition outcomes are allowed to mean*: local flatness, high estimated coverage, high confidence, apparent sufficiency or a provider failure cannot silently become proof of global completeness. This asymmetry is what permits the paper to absorb progress in retrieval without making its claimed authority boundary obsolete whenever a stronger retriever is published.

2.3 Empirical consequence

The decomposition also fixes the widening experiment. Candidate-generation methods should first be compared and composed under matched query/candidate budgets. Only after candidate recall is high enough to make stopping observable should the closure layer be challenged with direct ablations and strong stopping baselines. A positive external result must therefore show both acquisition quality and the incremental effect of typed closure authority; a cheaper stop that hides reachable relevant evidence is not a positive result. Conversely, a stronger donor retriever that improves acquisition while leaving the closure constraints intact strengthens the system rather than weakening the claim.

3 Formal mechanics

This section fixes the objects the evaluation measures. Every definition below is implemented in the system under test rather than assumed: work identity and read state, route independence and coverage, and route control each correspond to a module of the implementation, named in Section 8. Where a definition is deliberately conservative we state the failure it is conservative against, because the direction of the bias is what makes the refusal a safety property rather than a preference.

3.1 Work identity and content identity

Let R be the set of raw identifiers a route can emit (DOIs, arXiv ids, OpenAlex ids, local ids, URLs). Canonicalization is a map $\varphi : R \rightarrow \mathcal{S} \times V$ sending a raw identifier to a scheme-qualified key and a rendition version, so that $\varphi(\text{arXiv:2301.00001v1})$ and $\varphi(\text{arXiv:2301.00001v2})$ share a key and differ only in V : a version is a rendition of one work, not a second work. A declared scheme is treated as a claim rather than a credential; an identifier that does not match the shape of the scheme it announces resolves to UNKNOWN instead of entering the graph as a key that can never be fetched.

A *work* is the alias-closure of canonical keys. Given identity records with alias sets $A_1, \dots, A_n$, works are the connected components of the graph on $\bigcup_i A_i$ that joins two keys sharing a record. Closure is computed as a projection over an append-only ledger rather than at write time, because an alias learned late can join two works recorded early; folding at write time would freeze the wrong partition.

Separately from work identity, each retrieved rendition carries a content digest c . Route overlap, deduplication and coverage are computed over content digests, never over retrieval identifiers: a

source that re-mints identifiers would otherwise make two routes appear disjoint when they retrieved the same paper, which inflates every coverage quantity built on their overlap.

3.2 Question-conditioned read state

A read receipt is a tuple (w, c, σ, f) : work, content digest, extraction schema version, and frame f — the scientific question the extraction was performed for. Given the receipt set $\mathcal{R}$, the read decision for a candidate (w, c, σ, f) is

$$\text{read}(w, c, \sigma, f) = \begin{cases} \text{UNSEEN} & \nexists r \in \mathcal{R} : w_r = w \\ \text{CONTENT_CHANGED} & \exists r : w_r = w, \nexists r : (w_r, c_r) = (w, c) \\ \text{NEW_SCHEMA} & \exists r : (w_r, c_r) = (w, c), \nexists r : (w_r, c_r, \sigma_r) = (w, c, \sigma) \\ \text{NEW_FRAME} & \exists r : (w_r, c_r, \sigma_r) = (w, c, \sigma), \nexists r : r = (w, c, \sigma, f) \\ \text{ALREADY_READ} & \exists r \in \mathcal{R} : r = (w, c, \sigma, f). \end{cases}$$

ALREADY_READ therefore requires agreement on all four coordinates. This yields the distinction the evaluation depends on: a repeat extraction with all four coordinates equal is *duplicate processing* and is waste, while a repeat extraction under a changed content digest, schema or frame is a *legitimate reread* and is required work. Collapsing the two — treating “this paper has been seen” as “this paper has been read for this question” — is how a system reports coverage of material it only skimmed for a different purpose.

3.3 Earned route independence

A route capture is a tuple $\rho = (k, b, d, C)$: route kind, backend identity, query-derivation process, and the set C of captured content digests. For two captures ρ_i, ρ_j the independence predicate is

$$\text{ind}(\rho_i, \rho_j) \iff k_i \neq k_j \wedge b_i \neq b_j \wedge d_i \neq d_j,$$

and the verdict otherwise names the shared cause (SAME_ROUTE_KIND, SHARED_BACKEND, SHARED_QUERY_DERIVATION). Independence is thus *earned* from the generation and capture process; a distinct route label carries no evidential weight.

Proposition 1 (Direction of the conservatism). *Positive dependence between capture occasions biases unseen-mass estimators downward. Admitting a dependent pair therefore produces false confidence that search is near exhaustion, whereas refusing an independent pair only forfeits a diagnostic. The predicate is accordingly biased toward refusal.*

3.4 Coverage diagnostics are never authority

For an independent pair with captures of size n_i, n_j and overlap m_{ij} , the two-source estimate is $\hat{N} = n_i n_j / m_{ij}$, undefined at $m_{ij} = 0$. Zero overlap places no upper bound on the population, so the system returns a refusal with reason rather than a large number that would read as a measurement. When several independent pairs yield estimates, the reported value is $\max_{ij} \hat{N}_{ij}$: the largest population implies the most still unfound, which is the direction that fails safe. Every coverage estimate is marked diagnostic-only, and adaptive search is explicitly not a sequence of independent capture occasions — following up on what was just found converts singletons into doubletons and deflates the estimator toward “nearly done”.

3.5 Route control state machine

Let a route hold attempts $a_1, \dots, a_t$, each with retrieved digests D_i , novel digests $N_i \subseteq D_i$, cost, failure signatures and a backend-exhaustion flag. Marginal novelty of an attempt is $|N_i|/|D_i|$, and the flat streak z is the number of trailing attempts that succeeded, reported no exhaustion and returned no novel digest. Under a frozen policy with thresholds $\theta_{\text{ref}} \leq \theta_{\text{sw}}$, the decision is taken in strict priority order:

$$\delta = \begin{cases} \text{STOP (budget)} & \text{remaining budget} \leq 0 \\ \text{STOP (exhausted)} & a_t \text{ reports backend exhaustion} \\ \text{SUSPEND} & a_t \text{ carries execution-failure signatures} \\ \text{SWITCH} & z \geq \theta_{\text{sw}} \wedge \text{alternatives exist} \\ \text{REFORMULATE} & z \geq \theta_{\text{ref}} \\ \text{CONTINUE} & \text{otherwise.} \end{cases}$$

Two separations in this table are load-bearing. First, backend exhaustion and transport failure are distinct states: the former is a statement about the index, the latter about the channel, and a system that merges them converts an outage into evidence of absence. The latter yields SUSPEND, a non-terminal state carrying an open obligation, not STOP. Second, each terminal decision carries typed reopen triggers (budget increase, backend recovery or refresh, material frame change, material finding on an alternative route); a route reopens exactly when an observed trigger intersects that set.

Proposition 2 (Route-stop non-certification). *No individual route decision certifies task saturation. In the implementation every route decision, STOP included, has only route-local authority and carries an explicit declaration that it does not certify task saturation. Consequently, a route stop cannot be used by itself as a task-completeness certificate. A composition of route outcomes may support task closure only through a separately specified ensemble predicate whose route universe, material-obligation registry, provider/observability/custody guards, resource state and reopen triggers were fixed before the outcome.*

Proof. The route decision alphabet contains no task-closure authority, so projection of any one route output cannot emit a valid task certificate. The controller’s task decision is instead produced by the separately typed closure map over the full ensemble state. If that map’s registered completeness and validity guards hold, a composition can certify closure; if they do not, composition of route-local stops alone supplies no missing authority. $\square$

Task closure is therefore a separate judgement over the route ensemble. It requires that every route in the prospectively registered route universe is in a terminal or explicitly reopenable state, that no suspended or unavailable route leaves an unmet obligation, and — by Proposition 2 — that no coverage diagnostic has been used as its authority. It also requires the registered provider, observability, custody and resource-validity guards to pass as one conjunction. An unavailable route remains an open coverage obligation and is reported as such; it is never converted into evidence that nothing further exists.

3.6 What the evaluation must therefore measure

The definitions above fix four measurable error families, each with a ground-truth condition that only a complete-gold world supplies: route-stop false positives (a route stopped while it could still

reach unfound relevant items), route-stop false negatives (a route continued past its exhaustion point), premature task closure (task closed while a gold item was still reachable within remaining budget), and duplicate processing versus legitimate reread as separated by the four read coordinates. Route overlap, per-route marginal contribution and earned-independence status are computed over content digests. Metrics whose denominator is the gold set are reported only where that denominator is complete; the conditions are stated per metric in the frozen measurement plan.

4 Methods

The design choices in this section were fixed before the result-bearing offline campaign. Post-outcome edits to this prose are limited to correcting execution status (for example, saying that an external benchmark has not yet been run) and do not change a hypothesis, task family, system, metric, budget, exclusion rule, margin or decision rule. Machine-readable artifacts govern rather than this narrative: a design freeze, a measurement plan, a statistical plan, an external-artifact access audit, the controlled-world freeze and a prospective execution binding. Each is archived, and Section 8 names where.

4.1 Three-valued outcomes

Every decision this paper measures is three-valued rather than binary. A task may be determined complete, determined incomplete, or left *undetermined*; a route may be determined exhausted, determined to have more to give, or left undetermined. “We could not determine this” is recorded, reported and counted separately from “we determined this is false”, and the two are never merged. The distinction is the point: a provider outage, a missing credential or a censored route produces an undetermined outcome, and an undetermined outcome is not evidence of absence. Throughout the paper an outcome described as *undetermined* is this third value, carried in the evidence record and in every downstream count, never silently resolved to a negative.

4.2 Prospective discipline

The protocol is frozen and versioned before execution. A result-bearing execution is admissible only after subject code, data, evaluator, system identities, budgets and repeat seeds are bound. For the completed offline companion, the run manifest records that no outcome was accessed before the freeze, together with the exact subject revision, content hashes, 14 system identifiers, three repeat seeds and host-owned resource limits. The bound subject revision passed the full test suite before the first aggregate outcome snapshot was recorded. Publication summaries and manuscript prose may be generated after outcomes, but they may only project the frozen records; they cannot change the scientific decision rules.

4.3 Evaluation environments

Three environments are distinguished because no single one supplies both a complete denominator and open-world realism. The first supplies the only full system comparison with a complete denominator; later bounded external probes have narrower acquisition or screening authority and are reported separately.

Offline complete-gold controlled index. Recall is only measurable where the relevant set can be enumerated. The controlled index is therefore fully synthetic and self-generated. Each topic

has a materialized gold set under the frozen generator, and tests recompute the world and tasks from the recorded seed and compare all committed hashes rather than trusting a hand-authored denominator. Discoverability is heterogeneous: relevant items are partitioned across lexical, semantic, citation, reformulation and restricted/provider routes; two route labels deliberately share a backend; duplicate locators share content identity; revisions share identity but change digest; extraction-question-shift cases require legitimate rereads; and unavailable-route cases distinguish transport censoring from absence. Budgets are below exhaustive probing, so route allocation and stopping remain observable variables.

The completed suite has 390 tasks over five case families, drawn from 78 topics over 1,210 documents, frozen at suite fingerprint `2f6936ba52fb`. Every candidate uses the same host-owned session. The session alone can access the world, meters queries and reads, records route stops and transport states, and evaluates claims after the candidate returns. The candidate receives only a public task view and label-free retrieved records; protected gold and the relevance rule are never passed through the system interface.

External benchmark tasks. AutoResearchBench provides Wide and Deep scientific-discovery tasks [25]. The access audit establishes a narrower fact than an external result: Wide’s official exact IoU/recall/precision path scores normalized arXiv identifiers without a credential, while its sampled maximum-IoU-at- k family is noisy because upstream sampling is unseeded. Deep’s official title-matching evaluator needs an OpenAI-compatible judge. The Wide scorer was exercised on synthetic scorer input to verify that credential independence; no matched Wide result against a baseline is archived in this revision, and the scorer is not claimed to be vendored here.

SAGE is retained as neighbouring work and as a lexical-baseline warning [9], but is not executed as if its published setting were available: the paper-linked repository is data-only at the audited revision and provides neither an official runner nor a licence. Any substitute corpus, runner or weighting rule would be a declared deviation rather than the SAGE system. MetaSyn’s separation of retrieval from screening is adopted as a measurement constraint [24]; its credential-free BM25 plus public-protocol screening probe subsequently completed on all 86 released test reviews and is reported as a bounded retrieval/screening result, not as the full P2 system. AgentSLR remains a protocol-driven systematic-review neighbor/baseline family rather than an official evaluator dependency [15].

Frozen external retrieval snapshots and live-provider campaign. The prospective external design freezes a gold-blind candidate pool per query before comparing systems, then runs a smaller predeclared live subsample. Pool coverage must be reported as a shared ceiling because a gold-blind pool can omit gold. Live requests are designed to retain URL/parameters, query derivation, timestamp, raw response and typed transport outcome. The capture machinery and manifest exist and have been exercised, but the final result-bearing external pool and live campaign have not been executed in this revision. Therefore no external cost, latency or recall result is inferred from the capture audit.

4.4 Systems under test

The implementation evaluated here is the ORION discovery layer; it is the artifact under test, and the rest of the paper calls it the *governed system* so that the claims read as claims about the mechanism rather than about a product. The frozen offline companion compares the governed system with six confirmatory comparators: a BM25 keyword retriever, a dense retriever, a sparse–dense

hybrid, a one-pass retrieval-augmented pipeline, a single-route agentic searcher, and a protocol-driven systematic-review comparator. A seventh comparator, an adaptive multi-route exploratory searcher, was kept explicitly exploratory and was not retrofitted into the frozen confirmatory baseline set. The registry that binds each of these to its frozen system identifier is archived with the campaign.

These controlled-index systems are mechanism comparators, not claims to replicate a production BM25 engine, a particular embedding model or an external agent implementation. They consume the public synthetic route interface under identical budgets so that the variable under test is route/read/stopping governance. Their relevance screen sees only the public question and retrieved bibliographic text. The authored synthetic text makes screening deliberately easy; the experiment is not presented as evidence about classifier quality.

Six frozen ablations remove exactly one governance mechanism or authorize one forbidden transition: route-independence checking, the question coordinate in the read ledger, route/task stop separation, unavailable-route open state, coverage-diagnostic non-authority, and content-identity deduplication. All use the same session and budgets as the governed system.

4.5 Identity, reads and stopping

A read is keyed by content identity, content digest and extraction question in the full system. A new question can therefore reopen unchanged content for a legitimate reread, while mirrors with the same content identity/digest do not become new evidence. The corresponding ablations remove the question coordinate or replace content identity with document locator to expose suppression or duplicate work.

Route stop and task closure are separate actions. A route may stop after its public probe space is exhausted or after a typed transport outcome, but task closure is invalid if relevant material remains reachable within budget. An unavailable route stays censored and open rather than being converted to an empty result. Coverage diagnostics may inform a route policy but have no authority to certify global completeness.

4.6 Measurements

The offline evaluator reports denominator-valid complete-gold recall, precision, missed-gold count, route use, route/content overlap diagnostics, marginal relevant gain after the first route, premature task closure, censoring, duplicate processing and legitimate rereads, plus host-owned resource counters. Claims are normalized by content identity rather than locator so mirrors cannot inflate recall. Rich per-run artifacts retain route calls, reads, stop events, authority flags and raw-artifact hashes; the publication summary binds the full 16,380-record and rich-artifact sets by SHA-256.

The external measurement plan additionally defines official Wide/Deep scores, screening metrics and ranking-only metrics where their denominators and task semantics support them. Those definitions do not imply that an external result exists.

4.7 Statistics and authority

The statistical unit is the task. Repeated stochastic runs nest within task and system and never count as additional tasks. The frozen plan defines paired bootstrap comparisons, Wilson intervals for standalone binary rates, superiority/non-inferiority margins, a cost ceiling and Holm correction for the external confirmatory analysis. It also defines an authority ladder. The completed offline suite has $n = 390$, which reaches the precision tier the plan committed to in advance—registered in that plan as `TIER_B_committed` and requiring $n = 385$ —at an achieved half-width of 0.0496;

because that half-width exceeds the frozen superiority margin, the primary offline result carries the plan’s mandatory underpowered label. The three deterministic repeat seeds verify stable execution; they do not manufacture additional tasks. We therefore report descriptive absolute differences and mechanism diagnostics, with no promoted superiority decision from the offline companion.

For a future external confirmatory result, the frozen primary rule requires the family superiority margin of 0.03 under matched resources with the premature closure safety guard no worse than 0.02. Non-inferiority uses the predeclared 0.01 margin and is inadmissible above the 1.25 consumed-cost ratio. Secondary families use Holm correction. These rules remain prospective because no external system result is reported here.

4.8 Failures, exclusions and contamination

Candidate failures, malformed outputs, budget exhaustion, abstention and typed transport failures remain outcomes. Provider unavailability cannot become evidence of absence. The frozen evaluator uses an explicit failure taxonomy and terminal precedence; the publication table therefore warns that a lower-priority mechanism can be masked by a higher-priority terminal classification such as premature closure. Public benchmark questions and gold identifiers also create a contamination risk for future live search. A credential-free search-time self-exposure diagnostic over the frozen 400-task AutoResearchBench Wide candidate run is archived with the external results and reports a task rate of 0.0. That diagnostic does not claim absence of indirect, training-time, or live-provider contamination.

4.9 Decision rule

The paper-level rule is fail-closed. The offline companion can support a mechanism claim only at descriptive authority. External discovery superiority requires an archived denominator-valid external result under the frozen matched resources and safety rule. If that evidence is absent, blocked or undetermined, the claim is narrowed rather than promoted by analogy from the synthetic world.

5 Results

5.1 Authority of the result

Three result-bearing evidence families are complete in this revision, with different authority. First, the frozen offline complete-gold companion contains 390 tasks, 14 systems (the governed system, six confirmatory baselines, one explicitly exploratory comparator, and six ablations), and three predeclared repeat seeds, for 16,380 normalized run records. The systems are deterministic; the repeats therefore test harness stability and do not increase the statistical unit beyond 390 tasks. The precision tier the frozen statistical plan committed to in advance requires 385 tasks, which the assembled 390 tasks reach at an achieved half-width of 0.0496 for a standalone rate. Because that half-width still exceeds the frozen superiority margin of 0.03, the plan’s underpowered label attaches to the primary offline result: no superiority decision from the offline companion is promoted to a paper-level claim.

Second, the credential-free MetaSyn probe is a bounded external retrieval/screening result. It completed all 86 released test reviews using the pinned MetaSyn BM25 title+abstract retriever followed by the predeclared public-protocol screening rule, and was scored by the pinned official

ID-only evaluator. That probe is not the complete multi-provider system and is therefore reported separately from the paper’s still-open matched external comparison against a baseline.

Third, the archived public-screening transport campaign reports fixed-pool screening transport evidence on sources outside the SYNERGY development family. The one-pool Zenodo V1 binding failure, V2 post-schema-repair result and post-V2 active-comparator audit remain separate identities. A later SWIFT successor retained two additional population-mechanics failures and then bound five review decisions with 96,241 canonical rows. Its V3 unweighted review-level candidate-minus-u4 means were -0.021717971642 for recall at 10% screened and -0.050396333672 for WSS@95; u4 was better on both endpoints in all five reviews. The primary and relative-work-saving gates failed, while harm and absolute work saving passed without compensation. These screening studies inform downstream selection within acquired pools; they are not evidence about open-world route acquisition or closure, and V3 remains public-development evidence rather than population transport or confirmation. The subsequent KIFMS V6 packet is not a fourth performance family: it binds a lawful 14-review source, a 4,934-row content-disjoint population and a continuous recall-effort coprimary before exact V6 outcome access, but executes no arm. Its independent custody, label-semantic/class, harm, work-saving and performance coordinates therefore remain CANNOT-CHECK; V5 remains the latest independently unopened comparative screening result. V7 then uses a new hash-frozen transparent-public identity to execute those exact public data without claiming external custody. All 14 source files, 4,934 canonical rows and both-class checks pass. The learner/balancer means are $+0.084472$ CRE20, $+0.148820$ R@10 and $+0.119813$ WSS95, while the locked sign and harm gates fail; the full candidate loses to u4 at both R@10 and WSS95. V7 is therefore an adverse public-development mechanism result, not confirmation or promotion. V8 then tests whether a residual can safely envelop u4 using nested leave-one-review-out development. Exact u4 reproduces in 14/14 reviews, but 12/14 outer folds select fallback and both residual activations harm their held out review. Aggregate deltas versus u4 are -0.002009 CRE20, -0.028274 R@10 and -0.000991 WSS95, with worst R@10 -0.333333 ; no residual is admitted. V9 prospectively freezes a seven-review, source-disjoint title-emphasis test but stops before vectorization when one normalized title-abstract identity has two labels. V10 resolves only the identity semantics: provider records 1003 and 1018 have distinct author/DOI hashes and no provider duplicate link, so both are retained without choosing a label. Exactly one three-arm execution on 21,897 rows takes 357.789818 seconds. Candidate-minus-u4 means are $+0.000779993$ CRE20, -0.005382681 R@10 and $+0.001765467$ WSS95; positive CRE20 and R@10 occur in 4/7 and 1/7 reviews, while worst R@10 is -0.044444444 . The two coprimary magnitude gates and both sign gates fail, so the residual is discarded and exact u4 remains the fallback. Positive WSS95, harm and absolute-work-saving gates do not compensate for those failures.

V11 introduces no new empirical run. It proves a conditional safe-residual envelope: exact donor fallback is embedded, simultaneous noncompensatory certificates bound false admission at their joint noncoverage level, and finite V8/V10 failures do not identify universal u4 optimality. Strict same-contract separation from a saturated donor-complete class requires a fibre-separating signal or retained state, acquisition support outside the donor, or proof that the declared donor class was incomplete. A changed cost, access, world or authority contract changes the comparison rather than proving dominance under the original one.

V12–V15 then tested the first of those ascent routes without opening a performance outcome. V12 found different provider-native keywords on two records tied under exact u4 model text, but its normalization omitted V10’s trailing separator for an empty abstract. The exact V10 content-identity gate therefore failed, leaving the keyword difference as post-gate diagnosis only. V13 imported the exact V10 identity and passed all eight frozen bindings. It thus establishes one non-simulated provider-native coordinate that separates two records within an exact-u4-text fibre. Only

System	Recall	Premature closure	Pass	Cannot check	Fail
Governed system	0.979487	0.00	0.67	0.18	0.15
Protocol-driven SLR	0.666667	1.00	0.00	0.00	1.00
Sparse+dense hybrid	0.555556	1.00	0.00	0.00	1.00
One-pass RAG	0.555556	1.00	0.00	0.00	1.00
BM25/keyword	0.333333	1.00	0.00	0.00	1.00
Dense-route comparator	0.333333	1.00	0.00	0.00	1.00
Agentic single route	0.333333	1.00	0.00	0.00	1.00

Table 1: Controlled-index descriptive comparison. Rates are over 390 tasks; the failure column on the governed system’s row is honest budget exhaustion, not premature closure. The frozen plan’s underpowered label applies to the primary offline comparison, so no superiority interval or decision is reported here. A prospective external Wide campaign was executed but failed its frozen transport validity gate; its outcome is retained separately as undetermined.

four of the seven V10 reviews are keyword-capable, however, so no keyword-only residual can meet the unchanged six-of-seven sign gates and no performance run was authorized.

V14 sought a source-disjoint keyword-capable family but stopped at the source identity gate. The frozen commit resolves to a tree and index blob that are not the frozen expected blob; those expected bytes belong to a later commit, which was not substituted. All five identities are listed in the availability statement. V15 repaired the source-state binding without crossing into population or outcome access. The provider-valid signed commit, tag and recursive tree coherently expose the exact 22,135-byte index entry and 61 same-snapshot dataset CSV blobs. This is source-state availability, not an eligible census or screening benefit. The exact snapshot’s root licence is MIT, while current repository metadata reports CC0; these authorities are kept separate. The exact-index attestation endpoint returned 404, independent custody remained open, and zero index, census, ranking or metric run occurred.

The offline campaign is summarized by a single immutable publication record. It binds the full 16,380-record set by SHA-256 `c6430a651810f8e7a794aa0c1091794963c43389a1e5080c02c2807a2fc2c574` and the corresponding rich-artifact hash list by `d851f168faaf50969198180ecc61a6ac361c72556ceeb992f011af307dd00c37`. The exact run manifest was committed before the first outcome snapshot. The mechanism projection used for Figures 4–5 is archived separately; it records the same two source digests and is regenerated from host-owned traces rather than copied from manuscript numbers. The MetaSyn result and its exact integer error decomposition are bound by two further archived evaluator records. Section 8 names every one of these artifacts and where it lives.

5.2 Controlled-index comparison

The governed system attains mean complete-gold recall 0.979487 and precision 1.0. It passes 260 of 390 tasks, fails on 59 tasks after honestly exhausting its route-call budgets without certifying completeness, and leaves 71 tasks undetermined because a route holding unretrieved relevant material becomes unavailable; it never certifies a censored task complete. Its premature-task-closure rate is 0. The strongest of the six protocol-frozen confirmatory baselines is the protocol-driven systematic-review comparator, with mean recall 0.666667. That comparator claims task completion on all 390 tasks while reachable gold remains, so its premature-closure rate is 1.0. The descriptive recall difference is therefore +0.31282, with a descriptive premature-closure difference of -1.0. These numbers are evidence about the authored controlled world, not an estimate of real-web superiority.

The two systems are identical through their first three search calls in this authored world. The

protocol-driven comparator then reaches mean recall 0.666667 by query 7 and remains at that value through query 12. The governed system is at 0.757835 by query 7 and continues to gain, reaching 0.979487 by query 12. Figure 3 therefore localizes the descriptive gap to evidence gained after the early common search prefix rather than to an initial retrieval advantage.

5.3 Mechanism ablations

The safety ablations fail in the directions the frozen design intended them to probe. Removing the route-independence check reduces mean recall to 0.444444 and produces premature closure on every task. Allowing a route stop to certify task closure yields recall 0.333333 with the same 100% premature-closure rate. Letting a coverage diagnostic acquire stopping authority yields recall 0.555556 and again closes every task prematurely. These are negative controls: their purpose is to demonstrate that the forbidden transitions are observable, not to serve as competitive systems.

Removing content-identity deduplication is less catastrophic but measurably wasteful in the controlled world: mean duplicate-processing rate rises from 0 to 0.31453, 193 of 390 tasks exhaust the read budget, pass rate falls from 0.666667 to 0.369231, and mean recall falls to 0.939031. Removing the question coordinate from the read ledger leaves aggregate recall unchanged at 0.979487 but changes the extraction-question-shift mechanism exactly where expected: the mean count of legitimate rereads on that family falls from 3.666667 to 1.0. This is a mechanism result, not evidence of a recall gain from the ledger. Finally, removing the unavailable-route open state leaves mean recall numerically unchanged but turns the 71 censored undetermined cases into asserted completions and premature-closure failures. The distinction is therefore one of epistemic safety rather than headline recall.

The lexical route contributes 3.0 first relevant identities per task on average. Semantic and citation routes each contribute 2.0 additional identities, reformulation contributes 0.848718, and the restricted route contributes 0.966667; the last mean is below one because tasks that end censored or budget-exhausted before the restricted route is consulted contribute zero. Contribution and independence are deliberately separate variables: reformulation adds relevant content but shares the lexical backend and therefore cannot be promoted to an independent capture occasion.

Figure 5 exposes why numerical disjointness is not equated with route independence. In particular, lexical and reformulation have zero mean content overlap here yet are still rejected as independent because they share a backend. Conversely, lexical and semantic have nonzero mean overlap (0.07557) but remain structurally independent because neither backend nor query derivation is shared. The independence decision is therefore made from capture construction, while overlap remains a measured diagnostic.

5.4 MetaSyn external retrieval and screening probe

The corrected MetaSyn probe completed all 86 released test reviews with no failed review. Under the pinned official ID-only evaluator, macro retrieval recall is 0.7485 and retrieval precision 0.0627. Conditional retention among retrieved reference items is 0.7224; final inclusion recall is 0.5651, inclusion precision 0.1426 and inclusion F1 0.2033. The evaluator reports post-retrieval loss 0.1834 and screening accuracy 0.7228. Mean candidate output is fixed at 200 retrieved and 60 included items per review by the declared probe budget.

The evaluator-side integer ledger makes the stage loss concrete. Across the 86 reviews there are 1,677 reference-article links. The top-200 retrieval pools contain 1,079 of those links, so 598 are retrieval false negatives. Of the 1,079 retrieved reference links, 736 survive the deterministic screen and 343 are lost post retrieval. The final included sets therefore miss 941 reference links in total.

Gold is loaded only on the evaluator side after candidate output is frozen, and fresh regeneration verifies every deterministic official per-review metric before emitting the counts. These results identify where the bounded probe fails; they do not establish superiority for the full system.

5.5 Failure analysis and external status

Under the evaluator’s fixed terminal-failure precedence, the exploratory adaptive comparator exposes 296 premature-closure and 16 present-but-missed classifications while retaining 78 cases as CANNOT.CHECK; the no-dedup ablation exposes 193 budget-exhaustion failures. The governed system’s only terminal failure classification is honest budget exhaustion on 59 tasks: 260 tasks pass, 71 are censored, and premature closure remains at zero. The full per-system taxonomy is archived beside the publication summary.

A gold-defined route-stop replay is archived as supplementary material. The governed system makes 13 route-stop false positives in 1,950 route-stop events (0.006667): those restricted-route cases still have a theoretically reachable gold identity when the route stops. These do not become task-level false closures. The unavailable-route obligation remains open, so the governed system leaves the task undetermined rather than asserting completeness. Among the 1,878 governed task-routes that reach the oracle exhaustion point, there are zero route-stop false negatives. The restricted route records 377 post-exhaustion attempts in total; no task takes the two or more further attempts required for a route-stop false negative. This null-on-FN result is reported rather than being silently omitted, and it demonstrates why route-stop error and task-stop safety must be measured separately.

The remaining external authorities are kept separate from the completed MetaSyn probe. AutoResearchBench Wide’s released scorer is deterministic and credential-free, while the official Deep metric uses the released LLM title judge. That judge has now executed for the declared keyless public-arXiv probe: all 600 released Deep tasks were scored under the pinned judge prompt and model reached through a local protocol adapter, and the judge matched the reference title on 0 of them, a hit rate of 0.000. A nine-case positive/negative judge control—verbatim, truncated, and case-variant matches alongside distinct-paper non-matches—passed in full before the archive was accepted, so a zero-match outcome is attributable to the probe rather than to an always-no-match judge failure. This is a bounded external probe, not a matched Deep comparison against a baseline, and no Deep superiority claim is drawn from it. SAGE still lacks both the published retrieval corpus and an official evaluator. The frozen three-route live campaign also remains undetermined: its arXiv route is credential-free, but its two OpenAlex-backed route families require a funded OpenAlex API key, which the verified same-repository preflight did not find. No absent credential is converted into a negative outcome or a zero-cost measurement.

That zero was then diagnosed rather than left standing. A judge-free stage attribution over the same archive compared all 10,850 returned candidate titles against the 564 tasks that carried both a reference and candidates: exact-title recovery 0, substring recovery 0, and token-overlap recovery at a half-token threshold on 8 of them. Because a plain string matcher recovers almost nothing from the candidate sets themselves, the failing stage is candidate *generation* and not scoring, judging, or reference formatting. The attribution names a specific mechanism: questions whose incidental document-apparatus vocabulary—*supplementary*, *appendix*, *figure*—is treated by the acquisition mechanic as content vocabulary, so a title such as *Supplementary Orbit* outranks the paper the requested number actually sits in.

A successor acquisition mechanic was then built against that named mechanism and validated under a freeze written before any corpus was generated. The successor, a discriminative term-gating mechanic, uses index statistics only—never the gold set, the target, or the task family—

discarding question terms whose document frequency exceeds five percent of the corpus, ranking the survivors by inverse document frequency at the baseline’s own query width, and demoting rather than deleting documents that agree on fewer than two discriminative terms. On an authored corpus of 2,760 documents in which the named mechanism is present by construction and the denominator is complete, the current mechanic recovers the target for none of the 120 echo tasks within the top ten while the successor recovers all of them, at a rank-one recovery rate of 0.983. The paired McNemar test over those tasks is discordant on 120 pairs, every one of them favouring the successor. The harm guard is not vacuous: the current mechanic already scores 1.000 on the 60 tasks written without echo, and the successor does not damage them. BM25 over the same query recovers much of the top-ten rate (0.633) but almost none of the rank-one rate (0.0083), so the query-side gate rather than document-side weighting is what carries rank one. A pre-registered specificity control behaves as required: on 40 tasks whose question and target share no content token, the successor gains exactly 0.000, confirming the arena isolates the lexical mechanism instead of rewarding the successor generally.

This result is scoped to constructed reproduction only, and that scope was fixed before any number existed. The official probe cannot be re-run here: its frozen input is not in the repository and both public providers refuse this environment’s proxy. The successor is therefore evidence that the mechanism the attribution names is by itself sufficient to drive a lexical candidate generator to near-zero recovery, and that a discriminativeness-grounded generator is not driven to zero by it. It licenses no statement about target-hit rates on AutoResearchBench Deep, does not revive the archived probe, and does not discharge the paper’s successor-mechanic terminal, whose arena must be external and whose evaluator custody must sit outside the lane that authored the mechanic. One further limitation is declared in advance rather than discovered later: the successor requires corpus-level document frequencies, which are exact offline but against a live provider need a result-count endpoint or sampled index that this study does not supply.

A separate gold-blind OpenAIRE identity discriminator first established that structured arXiv persistent-identifier objects were available on 3 of 12 frozen public Wide questions, with 12 of 12 HTTP requests successful; that discriminator was forbidden to score or promote the paper. A subsequently frozen 400-row OpenAIRE/Crossref matched campaign then issued exactly 1,200 logical provider requests over one shared capture. The 400 Crossref discovery calls and 400 direct OpenAIRE calls succeeded, but every predeclared OpenAIRE DOI-crosswalk call returned HTTP 400. The observed provider-success fraction was therefore 0.666667, below the frozen 0.90 validity gate. The balanced baseline, direct-first diagnostic and governed projection were byte-identical, and their paired IoU differences were zero on all 399 distinct questions. The frozen analyzer correctly emitted an undetermined terminal, not a valid negative result.

Two further prospectively frozen campaigns followed, and together they correct the natural reading of that first one. A second campaign changed only the crosswalk encoding, from a single quoted OR expression to repeated persistent identifier parameters; its gold-blind transport probe returned HTTP 200 with an empty result array, below its prospectively frozen minimum of one, so acquisition never began and no score was produced. A third campaign changed the encoding again, to the documented identifier-filter OR list, and this time the transport probe validated: its receipt and raw response hashes were bound into the capture manifest and independently revalidated at score time. All 400 rows were acquired and scored. Provider validity reached 0.715833, again below the frozen 0.90, so the analyzer again emitted an undetermined terminal rather than a negative result.

The correction is this. The three systems were byte-identical in the first campaign, and their evaluator outputs were byte-identical again in the third, where the cross-backend route *was* valid and the three candidate files were genuinely distinct. So the identity is not explained by the failed

crosswalk. It is explained by the measurement floor. The frozen decision rule requires an average-IoU difference of 0.03 before superiority is supported, and the best-scoring arm in both completed campaigns achieved an average IoU of 0.003924 and 0.004374 respectively — roughly a seventh of the difference the campaign exists to detect. Deleting a competing arm outright would not reach the threshold. Three systems that all retrieve almost nothing tie on nearly every question, so their aggregate scores coincide and their scored outputs are identical files.

That matters for how the accompanying interval should be read. Both campaigns report a paired bootstrap interval of $[0.0, 0.0]$ over 399 matched questions, with 399 ties and no wins or losses, from ten thousand resamples, at the same confidence level and percentile method the campaign froze. Taken at face value that is the strongest equivalence evidence a bootstrap can produce. It is not evidence of equivalence at all: resampling 399 identical differences returns a zero-width interval at any sample size, and the interval would appear exactly this conclusive with four questions rather than four hundred. We therefore report no equivalence or non-inferiority conclusion from either campaign. Three reporting defects are recorded with them: both artifacts publish the unseeded sampled maximum-IoU-at- k family that this paper’s own scorer excludes by rule, that family decreases in k — which a maximum over the top k cannot do — and every runtime total is reported as exactly zero, which is untransported measurement rather than measured cost.

The terminals, workflow receipts and expiring Actions archives are retained in the repository, with checksummed mirrors beside them and a separate resolution audit; Section 8 names each location. A further campaign would require a new prospective freeze; none of these captures may be reinterpreted after the fact. The prerequisite is no longer a transport encoding. It is finding out why retrieval scores near zero on a benchmark whose gold is reachable, because a fourth transport version scored at the same floor would spend the same effort to learn the same nothing.

Therefore the externally supported discovery-benefit claim for the full system remains undetermined. The controlled-index result supports the narrower statement that the frozen governance mechanisms behave as designed under a complete denominator and matched host-owned budgets; the MetaSyn result supports only its declared external retrieval/screening probe. Bounded AutoResearchBench probe results, when archived, must remain separately labeled unless they exercise the final matched system under the frozen promotion plan.

6 Related work

Scientific-discovery benchmarks show that both targeted and open-ended literature search remain difficult for current agents [25, 10]. Acquisition methods are already heterogeneous. Strong lexical retrieval can outperform large-model retrievers inside deep-research workflows [9]; reasoning traces can become retrieval signals [3]; fielded admission and selective section fetching can improve the accuracy–context trade-off [23]; and bibliography expansion can recover studies missed by API-only search [20]. These methods enlarge the acquisition layer. They do not, without an additional argument, establish task completeness.

Automated evidence synthesis provides a second neighboring literature. MetaSyn separates retrieval from eligibility screening, making stage-specific false negatives visible [24]. AgentSLR evaluates an end-to-end systematic-review workflow over expert-annotated review tasks [15]. Technology-assisted review has long studied active screening protocols, autonomy, recall targets, and workload reduction [5, 6, 21]. The present work adopts stage attribution and fair-budget comparison, while focusing on the authority transition between a local acquisition outcome and a global closure claim.

Stopping research includes statistical stopping for systematic review [1, 12, 27], decision-theoretic continuation [8, 26], verification-aware completion [18, 4], conformal coverage for adaptive

reasoning [29], and structured sufficiency or gap judgments [7, 13]. Finite obligation certificates and evidence-bound workflow transitions offer still stronger donor mechanisms [2, 28]. We grant these mechanisms to the acquisition policy and ask a residual question: when a material route is unavailable, censored, or provider-invalid, may it disappear from the task-closure denominator? The exact-contract comparison is designed to change only that aggregation rule.

Capture-recapture methods and information foraging motivate, but do not settle, the same boundary. Capture-recapture estimates require defensible capture occasions and can be misleading under dependence [11, 19]. Information foraging describes the value of continuing or leaving a local information patch [16, 17]. In adaptive retrieval, however, a low-yield patch or a dependent route cannot be promoted into proof that the global literature has been exhausted. Question-conditioned memory likewise helps organize evidence [14], but the method here claims only the bookkeeping distinction between content identity and the question for which that content was processed.

7 Discussion and limitations

The controlled studies support a control-layer interpretation, not a retrieval leaderboard claim. In the complete-gold index, the governed controller maintains open obligations where the evaluator knows relevant material remains reachable. The ablations show that forbidden authority transitions are observable, while the exact-contract battery isolates the aggregation error made when unavailable material routes are excluded. These results demonstrate implementation behavior under authored contracts. They do not estimate real-web difficulty or prove that the declared route registry contains every material route.

The TREC-COVID result is the strongest external discriminator in the present paper because it tests the registered comparative claim on a public collection [22]. It is adverse. The recall noninferiority interval crosses the registered margin, and resource use moves in the wrong direction. The higher nDCG@10 is scientifically informative because it indicates better top-ranked relevance on many topics, but it answers a different question. A post-outcome substitution of nDCG for recall and cost would change the study rather than pass it. The paper therefore makes no external superiority claim.

Two separately frozen BEIR diagnostics sharpen the mechanism-level negative. Density normalization changes the nominal threshold but leaves every ArguAna decision identical to the original rule at all five depths. A subsequent query-conditional estimator again leaves all five ArguAna decisions unchanged, and SciFact is also unchanged; NFCorpus moves only slightly and remains well short of fusion. On ArguAna the rank-overlap marginal assigns near-zero expected contribution to every unread route at every query and prefix. The failure is therefore in the statistic’s information content, not only its threshold or query conditioning. The registered no-rescue rule forbids a fourth estimator under the same identity.

Several additional probes remain bounded or undetermined. A public MetaSyn probe quantifies retrieval and screening loss, but it does not exercise the complete controller. An AutoResearchBench Deep probe returns no official title matches after a successful judge control, which is a bounded negative diagnostic rather than a matched comparison. OpenAIRE/Crossref campaigns fail their provider-validity criterion; their zero paired differences cannot be read as equivalence. Public screening studies sometimes favor the frozen donor over the candidate. The result-retention ledger keeps these outcomes visible without averaging incompatible authority levels.

The formal control rules also have limits. Route independence is conservative and provenance-based; it does not prove statistical independence. The closure decision is relative to declared material routes and reopen conditions. A previously unknown source family can therefore reopen

a task. Content hashes support deduplication but do not decide scientific relevance. Finally, the controlled index supplies complete denominators by construction. Open-web completeness generally does not.

These limits shape the intended use. The control layer can sit beneath stronger retrievers, screening models, and stopping policies, because it constrains what their outputs mean rather than how they are produced. Its practical value is to prevent provider failure, local flatness, high estimated coverage, or apparent sufficiency from becoming an unsupported global claim. Whether this conservatism improves real scientific decisions at acceptable cost requires a new prospectively frozen external study.

8 Data and code availability

The implementation repository contains the controlled index, host-owned evaluation harness, the frozen offline systems and ablations, the descriptive analysis code, and the publication-layer regeneration commands. Every artifact named in the narrative is listed here with its path, so that no repository location has to be carried in the prose.

Preregistration

The study was preregistered and frozen before any outcome was accessed, under the identifier `P2.open-world-discovery.v1`. A reader checking whether a reported analysis was planned or chosen after the fact should compare against that freeze; it is the reason every number below can be traced to a decision made in advance of seeing it.

Frozen design and measurement rules

The design freeze is `protocol/PROTOCOL_V1.json`; the measurement and statistical rules are `protocol/MEASUREMENT_PLAN_V1.md` and `protocol/STATISTICAL_PLAN_V1.json`; external-artifact access states are `protocol/EXTERNAL_ACCESS_AUDIT_V1.json`; and the prospective execution binding for the result-bearing offline campaign, including the exact subject revision and Git blob identities, is `protocol/OFFLINE_RUN_MANIFEST_V1.json`, which also binds the frozen identifier of every system under test. The identifiers are `orion_full` for the governed system; `bm25_keyword`, `dense_retrieval`, `sparse_dense_hybrid`, `one_pass_rag`, `agentic_single_route` and `protocol_driven_systematic_review` for the six confirmatory comparators; and `adaptive_multiroute_exploratory` for the explicitly exploratory one. Each is a key of the systems map in `evidence/offline_results/RESULTS_SUMMARY_V1.json`, alongside the six ablation identifiers. The mapping from every result-bearing sentence in this manuscript to the archived artifact key that supports it is `protocol/CLAIM_LEDGER_V1.json`, with a human-readable view at `evidence/CLAIM_LEDGER_V1.md`.

The definitions in Section 3 are implemented in `src/orion/knowledge/`: work identity and read state in `identity.py` and `ledger.py`, route independence and coverage in `routes.py`, and route control in `route_control.py`.

Offline complete-gold companion

The frozen complete-gold world is distributed under `evidence/offline_gold/`; it is fully synthetic, regenerable from the recorded seed, and verified on every clean CI run against its committed content hashes and suite fingerprint.

The compact publication result is `evidence/offline_results/RESULTS_SUMMARY_V1.json`, and the mechanism projection behind Figures 4–5 is `evidence/offline_results/OFFLINE_MECHANISMS_V1.json`. The full per-system failure taxonomy is `evidence/offline_results/TABLE_P2-3_failure_taxonomy.md` and the gold-defined route-stop replay is `evidence/offline_results/TABLE_P2-S1_route_stop_oracle.md`. The command `python3 papers/orion-12-open-world-scientific-discovery/scripts/run_offline_companion.py --check` reconstructs all 16,380 runs and refuses drift. With `--write-raw DIR` it also emits the complete normalized JSONL result set and all rich run artifacts. The committed summary records SHA-256 digests of both complete sets, so a reproduction can be checked without requiring hundreds of generated files in the repository.

External benchmark artifacts

External benchmark artifacts are not redistributed. Their licences, access states and pinned revisions are recorded per artifact in `protocol/EXTERNAL_ACCESS_AUDIT_V1.json`, with source checks retained under `evidence/access/` and summarized in `protocol/TABLE_P2-1_freeze_manifest.md`. AutoResearchBench is pinned to upstream commit `a46c9bfb8968786f73f0a6a5b365b5384cd0f96d`; the released 1,000-record bundle is split by its explicit task-type field into 600 Deep and 400 Wide records before candidate custody. The Wide scorer is deterministic and credential-free. Deep official scoring uses the released title-identity judge and still requires an OpenAI-compatible endpoint; the repository workflow now records this authority precondition without exposing credential values. The credential-free search-time self-exposure diagnostic over the frozen Wide candidate run is `evidence/external_results/AUTORESEARCHBENCH_SEARCH_CONTAMINATION_V1.json`.

The completed MetaSyn probe is archived as `evidence/external_results/METASYN_ID_0_NLY_PROBE_V1.json`. It binds all 86 released test reviews, the pinned upstream revision, dataset revision, candidate-output tree and official ID-only evaluator output. The evaluator-side integer error ledger is separately archived as `METASYN_SCREENING_FN_LEDGER_V1.json`; its regeneration reloads the pinned gold only after candidate output is frozen and verifies the official per-review metrics before emitting exact retrieval and screening false-negative counts. These artifacts make the bounded MetaSyn retrieval/screening result reproducible without reclassifying it as a result for the full system.

The public screening transport successor is archived under `development/p2-zenodo-transport-successor-2026-08-23/`. The exact source identity and licence are in `SOURCE_FREEZE.json`; the accidental first-row label exposure is preserved in `SCHEMA_PREFLIGHT_INCIDENT.md`; and the failed V1 binding result remains `RESULT_V1_RETAINED.json`. The post-schema repair is frozen separately in `PROTOCOL_FREEZE_V2.json` and `IMPLEMENTATION_FREEZE_V2.json`; the componentwise estimand and non-transported coordinates are explicit in `P2_ONE_POOL_TRANSPORT_ESTIMAND_AMENDMENT_V1.json`. The executable ordering is `run_transport_successor_v2.py`, and the result and human-readable boundary are `RESULT_V2.json` and `RESULT_REPORT_V2.md`. The public source body is not committed; it is retrievable as Zenodo record 10423427 under CC BY 4.0 and is accepted only at the frozen byte count and MD5.

The post-V2 active-comparator audit is archived separately under `development/p2-zenodo-active-comparator-audit-2026-08-23/`. Its protocol and implementation were frozen before either comparator outcome; the pinned ASReview archive and selected donor source files are hash-bound in `PROTOCOL_FREEZE_V1.json`. The cadence-matched runner and immutable negative receipt are `run_active_comparator_audit_v1.py` and `RESULT_V1.json`. The audit does not redistribute the Zenodo body or the pinned donor archive and does not claim an exact ASReview

application replay.

The cross-review SWIFT V1–V4 records are archived under `development/p2-swift-cross-review-controller-transport-2026-08-23/`. The retained population-mechanics failures are `RESULT_V1.json`, `RESULT_V2.json`, `POPULATION_BINDING_FAILURE_DIAGNOSIS_V1.json` and `POPULATION_BINDING_FAILURE_DIAGNOSIS_V2.json`. V3 is bound by `PROTOCOL_FREEZE_V3.json`, `IMPLEMENTATION_FREEZE_V3.json`, `BINDING_RECEIPT_V3.json` and `RESULT_V3.json`; its human report and failure provenance are `RESULT_REPORT_V3.md` and `FAILURE_ATLAS_V3.json`. The post-outcome diagnostic-only freeze is `NEXT_CONTROLLER_SUCCESSOR_PROTOCOL_V4.json`. The local PubMed title/abstract snapshot is not redistributed: it is 90,520,080 bytes with SHA-256 962aeb56416dc72fe54336eb4867d14e17505b9b28cb7baf2fd469e82439a568. `PUBMED_SNAPSHOT_RECEIPT_V1.json` retains its acquisition receipt, while `SOURCE_PAYLOAD_MANIFEST.json` records the exact hashes and byte counts of all omitted source payloads. The V1–V3 frozen runners and original source payloads remain hash-bound but are not redistributed with the manuscript integration packet.

The content-disjoint SYNERGY V5 successor is archived under `development/p2-swift-content-disjoint-v5-2026-08-23/`. Its before-outcome ordering receipt, population/content freeze, protocol and implementation freezes, result, failure atlas and independent arithmetic/hash audit are `PREREGISTRATION_BEFORE_OUTCOME_RECEIPT_V5.json`, `POPULATION_AND_CONTENT_FREEZE_V5.json`, `PROTOCOL_FREEZE_V5.json`, `IMPLEMENTATION_FREEZE_V5.json`, `RESULT_V5.json`, `FAILURE_ATLAS_V5.json`, and `SCIENTIFIC_VERIFICATION_V5.json`. The 12 publisher files (287,754,769 bytes) matched their publisher SHA-1 and local SHA-256 values but are not redistributed. No title, abstract, label or active ordering is stored in the integration packet.

The outcome-unopened KIFMS V6 source/protocol packet is archived under `development/p2-continuous-recall-effort-v6-2026-08-23/`. The source rights and exact 14-file revision-one identity are `SOURCE_RIGHTS_PROVENANCE_RECEIPT_V6.json`; the label-blind 5,074-to-4,934 population calculation and its one raw V5 match are `LABEL_BLIND_OVERLAP_RECEIPT_V6.json`; and the population and continuous estimand freezes are `SOURCE_FAMILY_AND_POPULATION_FREEZE_V6.json` and `PROTOCOL_FREEZE_V6.json`. The feasibility result, negative ledger and integrity receipt are `SOURCE_FEASIBILITY_RESULT_V6.json`, `NEGATIVE_RESULT_LEDGER_V6.json` and `SCIENTIFIC_VERIFICATION_V6.json`. The final packet manifest has SHA-256 `cb7aac3ea6cc4b1070515dff882c4e3cb0a85632e5829af379208dd70448f710`; the protocol has SHA-256 `55a5363e6f9360c4722979d0cb11751575f050ce2eea2147a7883d679ea09633`. The 14 CC-BY-4.0 source CSVs and comparison source bodies are retrievable at their receipt-bound interfaces but are not redistributed in this packet. All KIFMS PubMed-identifier values are empty, so content hashes, not PMID matches, establish the final disjoint population. Historical public ASReview simulation artifact paths exist for these datasets; their metric contents and all exact V6 labels, classes, seeds, rankings and comparative scores remained unopened. Consequently the packet is not evidence of a globally unstudied family, independent custody or comparative performance.

The subsequent transparent public-development execution is archived under `development/p2-kifms-transparent-execution-v7-2026-08-23/`. Its protocol and implementation hashes were frozen before any KIFMS label, class count, seed, order or V7 metric was opened. `RESULT_V7.json` retains all four complete rankings per review through hashes and metrics but no title, abstract or row-level label; `RESULT_REPORT_V7.md` and `NEGATIVE_RESULT_LEDGER_V7.json` give the scientific diagnosis, and the packet-native validator checks the fixed hashes, 4,934-row population, both classes in 14/14 reviews, gate arithmetic and 14-file manifest. Exact public source CSVs remain outside the repository. V7 explicitly records `independent_custody=false`; it is not confirmation.

The nested donor-envelopment development successor is archived under `development/p2-donor-envelopment-v8-2026-08-23/`. The protocol, implementation freeze, exact result, adverse

scientific adjudication, recursive ledger and next discriminator are retained separately because the raw execution selector activated two residuals whereas the cross-fitted scientific result admits none. `SCIENTIFIC\_ADJUDICATION\_V8.json` is authoritative for the adverse terminal; the packet validator checks 14 exact u4 reproductions and 84 residual executions. No public source text or row-level label is redistributed.

The source-disjoint title-emphasis stop and its provider-native identity successor are archived under `development/p2-source-disjoint-title-emphasis-v9-2026-08-23/` and `development/p2-title-emphasis-conflict-resolved-v10-2026-08-23/`. V9 retains the single conflicting-label population stop and executes no model arm. V10 retains both provider-native nonduplicate records, executes the frozen three arms once on 21,897 rows, and records the failed coprimary magnitude and sign gates. Its result SHA-256 is `c69e5634b8d0e82a1fa393dbae276d728e9a2f7ba4c9db81a27c0329e2e66742`. No source text or row-level labels are redistributed, and neither packet is independent confirmation.

The conditional safe-residual and donor-saturation theory is archived under `development/p2-safe-residual-envelope-theory-v11-2026-08-23/`. The theory, finite witnesses, JSON claim ledger and receipt are bound by its SHA256SUMS; `THEORY.md` SHA-256 is `a3c9107ad152766efc2a08d722657d550781f1f181f47f3ceeba041c0a710176`. V11 performs no empirical or model run and does not establish that an unseen positive family exists.

The source-state successors are archived under `development/p2-state-expanding-acquisition-v12-2026-08-24/` through `development/p2-state-expanding-acquisition-v15-2026-08-24/`. Their SHA256SUMS files have SHA-256 values `f2a5cb0fb214d84807ee0396808edb36b619b59ca2066900a6026e163b98cef0`, `e01b9e6bf2e2169d318455a95e1b642a403e228b159eadb7d70295cb73e2c86e`, `8bb23e1c91d59328b64c70fc04af39840c78830f4d4cea1d588f2698495234e6`, and `c8bcfeafea18b06d58ab740da508a6d323de1a1b1ce946e43e0f79df1b114047`, respectively. The corresponding result SHA-256 values are `99c054ac292a7ceb2602323ca011f651f8f3ca3f00fb1a0d25b2684979836ff4`, `8f8681febfccc390f8e3392d321e01737c0ef4cfd2f4f3db45ab8410a15f2cbf`, `95496f053d762173285b9fbb436ff19bad08f47a82e278bb44715b565a0cd333`, and `2c394297445ad68019e7f559697d4141f5930532aa6dfef2e1525687bfb07107`.

V12 retains the failed exact-V10-identity gate and its post-gate diagnosis; V13 retains the positive two-record non-simulation witness and four-of-seven support ceiling. V14 records the frozen-commit index mismatch and the decision not to substitute later commit `dc2dadfdbb98eb1b4259604789abd640aa3b693e`. V15 binds the signed commit/tree/tag, 61-blob dataset manifest and exact historical MIT licence while keeping current CC0 metadata separate. Its unexecuted template error and the successor-only restored V14 route are recorded in `IMPLEMENTATION_CORRECTION_V15B.json`, SHA-256 `55dda1c29a0e0a619f7234888ffdf0d20a2e35d4b8782972343254d694607980`. The outstanding three-role custody handoff is `INDEPENDENT_SOURCE_CUSTODY_REQUEST_V15.json`, SHA-256 `f07c8185c48d5e0a40d02e7bee4dcc0530922fe68e111c0a4cbd83c7c36e2a58`. No packet contains an index parse, review census, model ranking or performance metric that can promote the manuscript terminal.

The OpenAlex V1 development trace received a rights-preserving archival redaction after an adversarial audit found decrypted `Q0_RAW` strings in two duplicate in-tree traces. Original bytes with SHA-256 `10ffd34a8b3de5e3b9e45b0e309bf451ff201c2dd0c5d1837e70504959d684cd` are quarantined outside the redistributed worktree at mode 0600. The in-tree copies omit the plaintext and retain only per-task SHA-256 provenance; their redacted SHA-256 is `3eae226dfb323ad812b67b533be70041345b734e54a447fddedf9f1c2881797`. The machine-readable mapping is `development/p2-openalex-public-successor-2026-08-23/Q0_RAW_QUARANTINE_RECEIPT_V1.json`. Authoritative upstream permission for plaintext redistribution remains `CANNOT-CHECK`; no plaintext circulation is authorized and this archival action is not a legal opinion.

All three OpenAIRE/Crossref Wide campaigns are preserved rather than left behind an expiring Actions URL. Their exact results and run receipts are stored under `evidence/external_results/`, one prospective freeze per campaign under `protocol/`; checksummed mirrors of the shared public candidate capture, the failed first evaluator handoff, and the successful evaluator-only repair are under `evidence/ci_mirror/`. The mirror manifest records both archive-level and per-file SHA-256 digests. Evaluator gold never entered any candidate capture. The first campaign is undetermined because 400 of 1,200 planned provider calls failed; the second because its transport probe returned an empty result array below its frozen minimum, so acquisition never began; the third because provider validity reached 0.716 against a frozen 0.90. None is undetermined because evidence was discarded.

The prospective freeze for the successor acquisition mechanic is `protocol/P2_LEXICAL_ECHO_SUCCESSOR_FREEZE_2026-08-21.json`, written before any corpus was generated; the mechanic itself is registered there under the identifier `D4_DISCRIMINATIVE_TERM_GATING`.

The resolution audit that establishes the measurement-floor finding is under `evidence/audit/`. It reads the published campaign results and reports whether each could have produced a different verdict; it is reproducible from the repository and carries, in the same artifact, a control campaign with resolution so that a check firing on every input can be told from a finding.

The live-provider capture implementation and campaign manifest retain request URL/parameters, query derivation, timestamp, raw body, hash and typed transport status. The frozen live protocol requires arXiv plus two OpenAlex-backed route families. A non-leaking Actions preflight is archived as `evidence/access/LIVE_PROVIDER_PREFLIGHT_2026-08-16.json`; it confirms that arXiv is credential-free but that no funded OpenAlex API key is available in the same-repository execution context. Therefore the three-route live campaign cannot be frozen for execution here, and no raw live campaign or live monetary-cost ledger is fabricated. Operator re-scope 2026-08-17 removed the empirical-cost requirement: the sanctioned cost axis is the host-recorded search-query count (Figure 3, `protocol/TABLE_P2-2_baseline_ablation_cost.md`), not a live OpenAlex or arXiv API bill.

SIEVE external-arm preflight

The exact candidate identity and terminal-preserving adapter requirements are recorded in `protocol/P2_SIEVE_ADAPTER_PREFLIGHT_V1.json`. They bind SIEVE/SkimSearchAgent to the paper-linked Apache-2.0 repository revision `4a640296770c39b8274e7678c6de844daea9541c` and prohibit its native answer, empty-answer, error or timeout states from acquiring P2 closure authority. The bounded one-instance stub fixture and its trace/result hashes are archived under `development/comparator-binding-harness-2026-08-23/`; they establish only that the pinned entrypoint can run. The current status is `P2_SIEVE_RUNNABLE_FIXTURE__NOT_A_FROZEN_ARM`: no archived P2 task-world reproduction, matched provider/route/budget binding, or comparator performance result is supplied by this preflight.

9 Reproducibility

Four levels are distinguished, and we claim only what the current artifacts support.

Fully reproducible without network access. The 390-task offline controlled-index companion regenerates from committed data and subject code. A clean GitHub Actions job independently rebuilds the 16,380-run publication projection and compares it to the committed summary. The frozen plan awards the result its committed precision tier together with a mandatory underpowered label on the primary offline comparison; reproduction does not elevate that authority.

Reproducible bounded external evidence. The pinned MetaSyn workflow reconstructs the public/gold split, runs the credential-free BM25 plus public-protocol screening candidate, executes the official ID-only evaluator, and can regenerate the exact false-negative ledger. This is external evidence for that declared probe only.

Runnable credential-free benchmark lanes. AutoResearchBench Wide scoring is deterministic once the gold-blind candidate output is frozen. The same public-arXiv candidate machinery also supports the declared deterministic Deep exact-ID deviation. The official Deep title judge remains a distinct authority and is not replaced by the ID deviation when no judge endpoint is available.

Executed official Deep judge archive. The bounded keyless public-arXiv probe of the official Deep LLM title judge is archived as `evidence/external_results/DEEP_OFFICIAL_ARCHIVE_V1.json` with its nine-case positive/negative judge control in `DEEP_JUDGE_CONTROL_2026-08-17.json`. The checkpointed judge runner and the archive finalizer are both reproducible from the repository as `scripts/judge_checkpointed.py` followed by `scripts/finalize_deep_scoring.py`; the archive records the judge model, the pinned prompt digest, the run manifest hashes, and the bounded probe scope label.

Capture-capable but blocked as a final live campaign. The committed live layer is designed to archive mutable provider evidence rather than re-derive it, but the frozen route set requires funded OpenAlex access that is absent in the verified execution context. Until that authority is supplied and the campaign is actually run, raw live-provider outcomes, empirical live monetary cost and wall-clock, and the matched external discovery claim all stay undetermined. Query-count metrics from the offline companion are the completed cost evidence in this revision.

10 Ethics

This work does not collect new human-subject records or personally identifiable information. The frozen complete-gold companion is fully synthetic and regenerable from a recorded seed. External probes use publicly released benchmark questions and evaluator interfaces; third-party corpora are not redistributed where the access-audit table records a licence block or partial redistribution. Dual-use risk is that of any scientific-literature retrieval system: the paper evaluates discovery governance and does not present a deployment, surveillance, or human-facing study.

11 Conclusion

Open-world scientific-literature discovery needs separate control of evidence acquisition and closure authority. The method developed here derives route identity from provenance, distinguishes content identity from question-conditioned processing, keeps route stopping local, and retains unresolved material routes as open obligations. Controlled and exact-contract tests show that these rules behave as designed. The registered TREC-COVID recall-and-cost gate nevertheless fails, and favorable nDCG does not replace that decision. The later density-normalized and query-conditional successors also leave every ArguAna stopping decision unchanged, showing that the frozen rank-overlap marginal is signal-inert there. The present contribution is therefore a bounded, reproducible control method for representing acquisition authority. External retrieval benefit and open-world completeness remain unestablished.

Declaration of generative AI and AI-assisted technologies

During preparation of this work, the authors used OpenAI ChatGPT to support literature searching, source checking, reproducibility review, manuscript editing, and submission-package preparation. The tool was not treated as an author and did not replace scientific judgment. Result-bearing statements and citations were checked against the archived evidence and primary-source records. The authors remain responsible for the article’s content.

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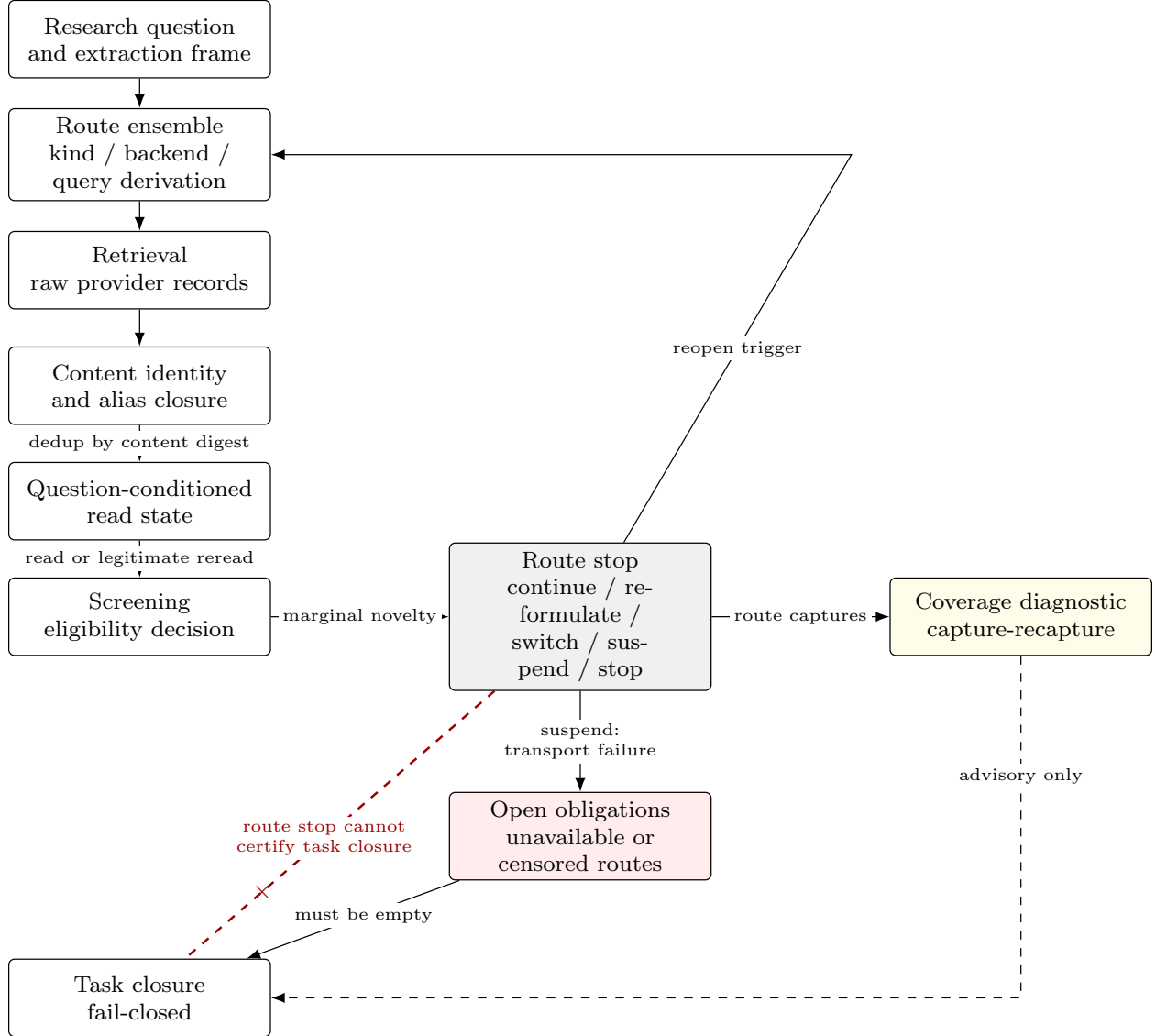


Figure 2: The discovery pipeline separates retrieval, screening, route stopping, and task closure. Solid arrows carry state updates. The dashed coverage diagnostic informs search but does not certify closure, and the crossed arrow is the forbidden promotion from a route stop to task completeness. Transport failure leaves an open obligation rather than evidence of absence.

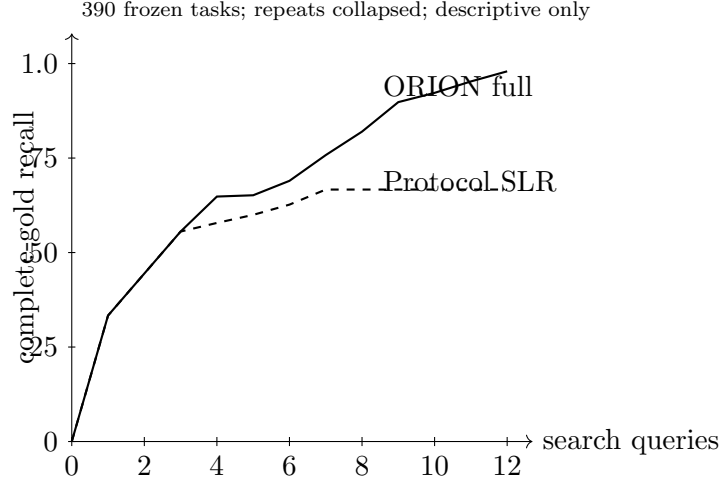


Figure 3: Cumulative complete-gold recall versus the host-recorded number of search queries for the governed system and the strongest frozen confirmatory baseline. Deterministic repeats are first checked for identical traces and then collapsed within task; the curves are means over the 390 tasks and are descriptive only. Synthetic wall-clock is intentionally fixed to zero, so the controlled companion supports the query axis, not an empirical-time claim.

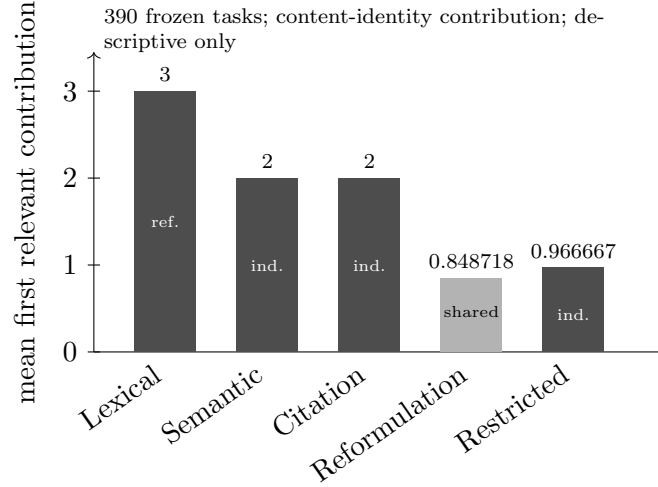


Figure 4: Mean number of first relevant content identities contributed by each route in the governed system’s campaign order. “Independent” is an earned structural annotation relative to the lexical reference route; the reformulation route is deliberately marked shared-backend even though it adds a new identity. Values are task means over the frozen descriptive companion.

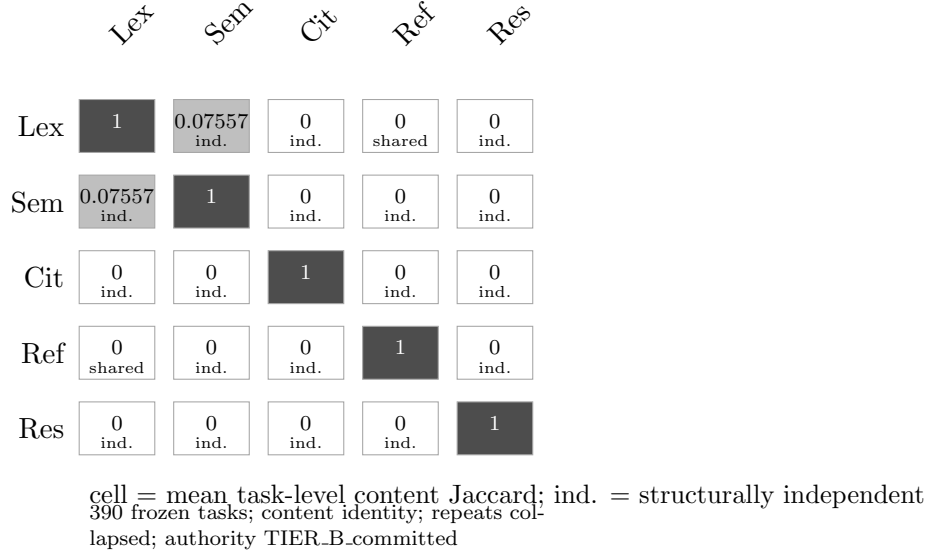


Figure 5: Mean task-level Jaccard overlap between route captures, computed over content identities rather than source locators, together with the structural independence verdict for each off-diagonal pair. The lexical–semantic overlap is 0.07557; the other off-diagonal mean overlaps are zero in this controlled world. Zero observed overlap is not itself evidence of independence.

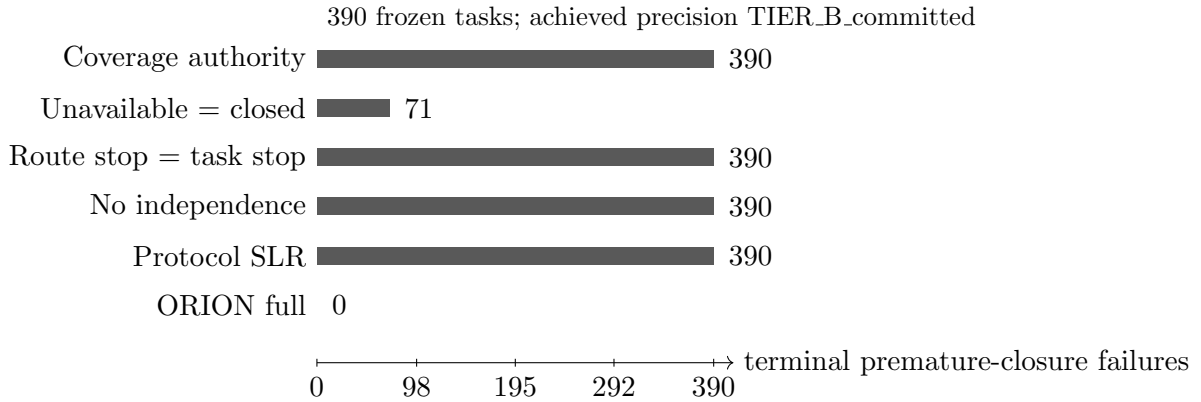


Figure 6: Terminal premature-closure classifications for selected stopping-safety controls in the frozen 390-task offline companion. The plot is generated from the immutable publication summary. It shows task counts, not an inferential estimate; lower-precedence failure mechanisms can be masked by the evaluator’s terminal-failure precedence.